Project: Worked examples
Title: Rolling bearing rating life
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Equivalent dynamic load of a rolling bearing under radial and axial load, its basic rating life L_{10} in millions of revolutions and in hours at the running speed, and the life modified for reliability, against a required life.

WHAT IT CALCULATES	RELATION
Equivalent dynamic load from the radial and axial loads	$P = X \cdot F_r + Y \cdot F_a$
Basic rating life in millions of revolutions (p = 3 for ball and 10/3 for roller bearings)	$L_{10} = (C/P)^p$
Rating life in hours, at speed n	$L_{10}h = L_{10} \cdot 10^6$ revolutions at speed n
Life adjusted for the required reliability	$L_{nm} = a_1 \cdot L_{10}$; $L_{nmh} = a_1 \cdot L_{10}h$
Range of validity of the life equation	$P \le 0.5 \cdot C$

Basis: Basic dynamic load rating and basic rating life[1](§6); Rolling-contact bearings[2](Ch. 11).

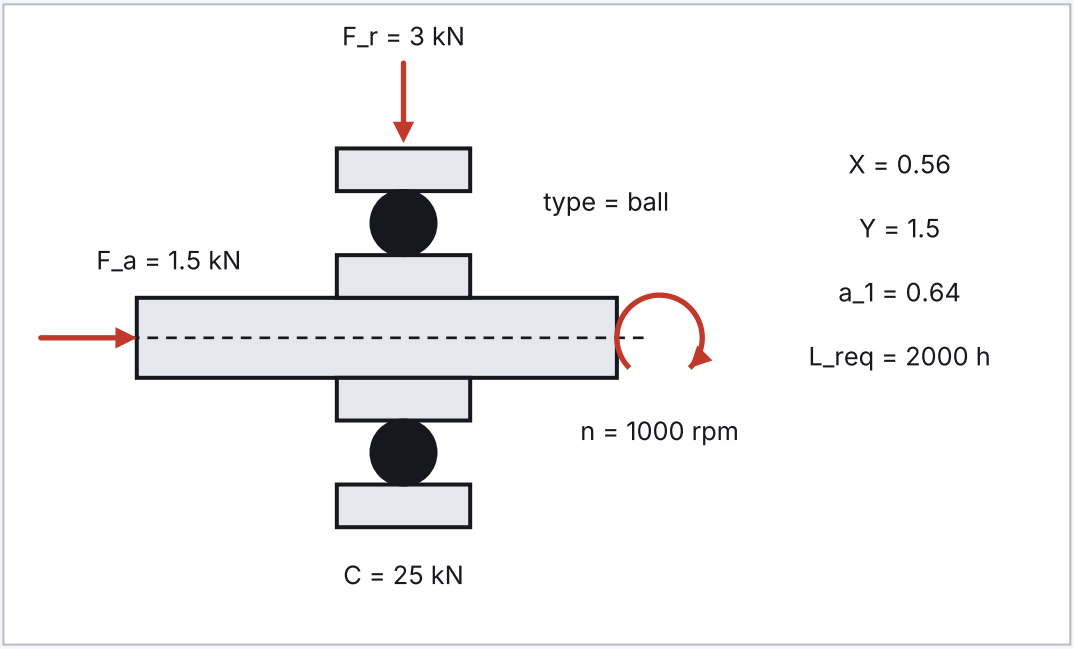


Figure 1 — A ball bearing on its shaft in section, radial and axial load, running speed n. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
type	Bearing type (selects the life exponent)	ball		drawing: DRG-GB-211 rev A	any
C	Basic dynamic load rating	25	kN	drawing: DRG-GB-211 rev A	0.1 … 100000 kN
F_r	Radial load	3	kN	spec: LOAD-SCHEDULE-01	0 … 100000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_a	Axial load	1.5	kN	spec: LOAD-SCHEDULE-01	0 … 100000 kN
X	Radial load factor (bearing catalogue)	0.56		drawing: DRG-GB-211 rev A	0 … 2
Y	Axial load factor (bearing catalogue)	1.5		drawing: DRG-GB-211 rev A	0 … 5
n	Speed	1000	rpm	spec: LOAD-SCHEDULE-01	1 … 100000 rpm
a_1	Life modification factor for reliability	0.64		spec: SPEC-MECH-03 design criteria	0.05 … 1
L_req	Required life (0 for no criterion)	2000	h	spec: SPEC-MECH-03 design criteria	0 … 1000000 h

3. Assumptions

A1. Steady load and speed; the bearing is correctly fitted, lubricated and free of contamination. (Basic rating life conditions.)
A2. X and Y are the catalogue factors for this bearing and load ratio. (Supplied by the user.)

4. Working

[10] p = 3 [1](§6)
= 3
Life exponent (ball bearing)

[11] P = X * F_r + Y * F_a [1](§6)
= 0.56 × 3 kN + 1.5 × 1.5 kN
= 3.93 kN
Equivalent dynamic load

[12] k_P = P / C
= 3.93 kN / 25 kN
= 0.1572
Load ratio P/C

[13] L_10 = (C / P)^p [1](§6)
= (25 kN / 3.93 kN)^3
= 257.42
Basic rating life (millions of revolutions)

[14] L_10h = L_10 * 1e6 * 2 * pi / n
= 257.42 × 1e6 × 2 × π / 1000 rpm
= 4290.3 h
Basic rating life in hours (10^6 revolutions of 2π rad at speed n)

[15] L_nm = a_1 * L_10
= 0.64 × 257.42
= 164.75
Modified rating life (millions of revolutions)

[16] L_nmh = a_1 * L_10h
= 0.64 × 4290.3 h
= 2745.8 h
Modified rating life in hours

[17] u_life = L_req / L_nmh
= 2000 h / 2745.8 h
= 0.72838
Required over modified life

5. Results

SYMBOL	VALUE	DESCRIPTION
P	3.93 kN	Equivalent dynamic load
k_P	0.1572	Load ratio
L_10	257.42	L_{10} (millions of revolutions)
L_10h	4290.3 h	L_{10} in hours
L_nm	164.75	Modified life (millions of revolutions)
L_nmh	2745.8 h	Modified life in hours

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Modified life against required life	u_life = 0.72838	u_life ≤ 1	0.728	[1](§6)

7. Applicability

No warnings.

8. Verification

Template mech.bearing-life@1.0.1 Content hash 888b0e414c8926cf Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. International Organization for Standardization. Rolling Bearings—Dynamic Load Ratings and Rating Life. ISO 281. ISO.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.